

Antagonistic effects of sulfide and butyrate on proliferation of colonic mucosa: a potential role for these agents in the pathogenesis of ulcerativ...

It has been shown that feces of patients with ulcerative colitis uniformly contain sulfate reducing bacteria. Sulfide produced by these bacteria interferes with butyrate-dependent energy metabolism of cultured colonocytes and may be involved in the pathogenesis of ulcerative colitis. Mucosal biopsies from the sigmoid rectum of 10 patients (no cancer, polyps, inflammatory bowel disease) were incubated with either NaCl, sodium hydrogen sulfide (1 mmol/L), a combination of both sodium hydrogen sulfide and butyrate (10 mmol/L), or butyrate. Mucosal proliferation was assessed by bromodeoxyuridine labeling of cells in S-phase. Compared to NaCl, sulfide increased the labeling of the entire crypt significantly, by 19% ($p < 0.05$). This effect was due to an expansion of the proliferative zone to the upper crypt (compartments 3-5), where the increase in proliferation was 54%. Sulfide-induced hyperproliferation was reversed when samples were coincubated with sulfide and butyrate. The study shows that sodium hydrogen sulfide induces mucosal hyperproliferation. Our data support a possible role of sulfide in the pathogenesis of UC and confirm the role of butyrate in the regulation of colonic proliferation and in the treatment of UC.